

Derivation of Equation of Motion of Space Manipulator System

1 Frame Transformation

Let B be the satellite base coordinate frame and N be the inertial coordinate frame. Any vector ${}^B\mathbf{r}$ expressed in the base frame can be transformed to the inertial frame as

$${}^N\mathbf{r} = C_{NB} {}^B\mathbf{r}, \quad (1)$$

where C_{NB} is the direction cosine matrix (DCM) defined as

$$C_{NB} = R_x(\alpha_b)R_y(\theta_b)R_z(\psi_b). \quad (2)$$

In Eq. (2), α_b , θ_b , and ψ_b are the base satellite's roll, pitch, and yaw angles, respectively. $R_x \in \mathbb{R}^{3 \times 3}$ is the rotation matrix about x -axis, $R_y \in \mathbb{R}^{3 \times 3}$ is the rotation matrix about y -axis, and $R_z \in \mathbb{R}^{3 \times 3}$ is the rotation matrix about z -axis. Note that the top-left superscript indicates the frame information where the elements of the vector are expressed. The inertia matrix of base satellite ${}^B I_b$ with respect to its center of mass can be transformed to the inertial frame as [3]

$${}^N I_b = C_{NB} {}^B I_b C_{NB}^T. \quad (3)$$

Unlike DCMs, which require 9 elements, quaternions use only 4 parameters to describe the attitude between two frames, reducing storage and computational cost. Quaternion-based transformations are preferred over Direction Cosine Matrices (DCMs) because they avoid singularities (e.g., gimbal lock) and are computationally efficient. A unit quaternion representing a rotation by a principal rotation angle θ about a principal rotation axis $\hat{\mathbf{u}} = [u_x; u_y; u_z]$ is [1]

$$\epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \end{bmatrix} = \begin{bmatrix} u_x \sin(\frac{\theta}{2}) \\ u_y \sin(\frac{\theta}{2}) \\ u_z \sin(\frac{\theta}{2}) \\ \cos(\frac{\theta}{2}) \end{bmatrix}. \quad (4)$$

Note that the quaternions satisfy the following constraints:

$$\epsilon^T \epsilon = 1. \quad (5)$$

The rotation matrix using quaternions is defined as [1]

$$R_\epsilon = \begin{bmatrix} 1 - 2(\epsilon_2^2 + \epsilon_3^2) & 2(\epsilon_1\epsilon_2 - \epsilon_4\epsilon_3) & 2(\epsilon_1\epsilon_3 + \epsilon_4\epsilon_2) \\ 2(\epsilon_1\epsilon_2 + \epsilon_4\epsilon_3) & 1 - 2(\epsilon_1^2 + \epsilon_3^2) & 2(\epsilon_2\epsilon_3 - \epsilon_4\epsilon_1) \\ 2(\epsilon_1\epsilon_3 - \epsilon_4\epsilon_2) & 2(\epsilon_2\epsilon_3 + \epsilon_4\epsilon_1) & 1 - 2(\epsilon_1^2 + \epsilon_2^2) \end{bmatrix} \quad (6)$$

Any vector ${}^B\mathbf{r}$ expressed in the base frame can be transformed to the inertial frame using the quaternion-based transformation matrix as

$${}^N\mathbf{r} = R_{\epsilon_{NB}} {}^B\mathbf{r} \quad (7)$$

2 Spacecraft Base Kinematics

Euler angles may suffer from the representation singularity issue, therefore, quaternions are considered to describe the attitude of the base spacecraft with respect to the inertial frame. The differential kinematic equation of the base spacecraft is defined as [1]

$$\dot{\epsilon} = \frac{1}{2}G(\epsilon)\omega_b, \quad (8)$$

where $\boldsymbol{\epsilon} = [\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4]^T \in \mathbb{R}^4$ is the quaternions and $\boldsymbol{\omega}_b \in \mathbb{R}^3$ is the angular velocity of the base spacecraft with respect to the inertial frame. Here, the first three elements (ϵ_1, ϵ_2 , and ϵ_3) are the vector component defined as $\boldsymbol{\epsilon}_v = [\epsilon_1, \epsilon_2, \epsilon_3]^T$ and ϵ_4 is the scalar component. Note that the quaternions must satisfy the holonomic constraints: $\boldsymbol{\epsilon}^T \boldsymbol{\epsilon} = 1$. The matrix $G(\boldsymbol{\epsilon}) \in \mathbb{R}^{4 \times 3}$ is defined as

$$G(\boldsymbol{\epsilon}) = \begin{bmatrix} \epsilon_4 & -\epsilon_3 & \epsilon_2 \\ \epsilon_3 & \epsilon_4 & -\epsilon_1 \\ -\epsilon_2 & \epsilon_1 & \epsilon_4 \\ -\epsilon_1 & -\epsilon_2 & -\epsilon_3 \end{bmatrix}. \quad (9)$$

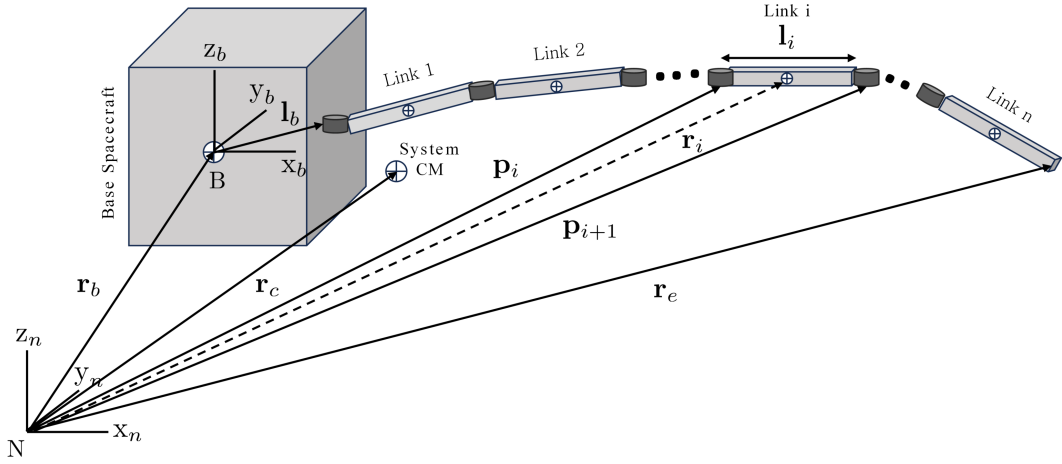


Figure 1: A free-floating Space Manipulator System

3 Deriving the Equation of Motion

The end effector position vector ${}^N \mathbf{r}_e \in \mathbb{R}^{3 \times 1}$ in the inertial frame is defined as

$${}^N \mathbf{r}_e = {}^N \mathbf{r}_b + {}^N \mathbf{l}_b + \sum_{i=1}^n {}^N \mathbf{l}_i \quad (10)$$

where ${}^N \mathbf{r}_b \in \mathbb{R}^{3 \times 1}$ is the position vector of the base centroid in the inertial space, ${}^N \mathbf{l}_b \in \mathbb{R}^{3 \times 1}$ is the vector of the base centroid pointing to the robot mounting point in the inertial frame, and ${}^N \mathbf{l}_i \in \mathbb{R}^{3 \times 1}$ is the vector between two adjacent joints (joint i and joint $i+1$) in the inertial frame. The vectors ${}^B \mathbf{l}_b \in \mathbb{R}^{3 \times 1}$ and ${}^i \mathbf{l}_i \in \mathbb{R}^{3 \times 1}$ are transformed from their corresponding frames to the inertial frame as

$${}^N \mathbf{l}_b = R_{\epsilon_{NB}} {}^B \mathbf{l}_b \quad (11)$$

and

$${}^N \mathbf{l}_i = R_{\epsilon_{Ni}} {}^i \mathbf{l}_i \quad (12)$$

Inserting Eqs. (11) and (12) into Eq. (10) results in

$${}^N \mathbf{r}_e = {}^N \mathbf{r}_b + R_{\epsilon_{NB}} {}^B \mathbf{l}_b + \sum_{i=1}^n R_{\epsilon_{Ni}} {}^i \mathbf{l}_i \quad (13)$$

Taking the time derivative of Eq. (13) with respect to the inertial frame yields

$${}^N\dot{\mathbf{r}}_e = {}^N\dot{\mathbf{r}}_b + \dot{R}_{\epsilon_{NB}} {}^B\mathbf{l}_b + \sum_{i=1}^n \dot{R}_{\epsilon_{Ni}} {}^i\mathbf{l}_i \quad (14)$$

The time derivative of the rotation matrix $R_{\epsilon_{Ni}}$ is given by [1, 2]

$$\dot{R}_{\epsilon_{Ni}} = \boldsymbol{\omega}_{i/N} \times R_{\epsilon_{Ni}} \quad (15)$$

where $\boldsymbol{\omega}_{i/N}$ is the angular velocity of body i with respect to inertial frame N . For the ease of representation $\boldsymbol{\omega}_{i/N}$ is represented as $\boldsymbol{\omega}_i$, and the superscript N from the vectors expressed in the inertial frame is dropped from the vector notations. Equation (14) can be re-written as

$$\dot{\mathbf{r}}_e = \dot{\mathbf{r}}_b + \boldsymbol{\omega}_b \times \mathbf{l}_b + \sum_{i=1}^n (\boldsymbol{\omega}_i \times \mathbf{l}_i) \quad (16)$$

where $\dot{\mathbf{r}}_b \in \mathbb{R}^{3 \times 1}$ is linear velocity of base centroid, $\boldsymbol{\omega}_b \in \mathbb{R}^{3 \times 1}$ is the angular velocity of base, and $\boldsymbol{\omega}_i \in \mathbb{R}^{3 \times 1}$ is the angular velocity of link i . Using the relation $\boldsymbol{\omega}_i = \boldsymbol{\omega}_b + \sum_{j=1}^i \dot{\mathbf{q}}_j \hat{\mathbf{k}}_j$, Eq. (16) can be rewritten as

$$\dot{\mathbf{r}}_e = \dot{\mathbf{r}}_b + \boldsymbol{\omega}_b \times (\mathbf{r}_e - \mathbf{r}_b) + \sum_{i=1}^n \dot{\mathbf{q}}_i \hat{\mathbf{k}}_i \times (\mathbf{r}_e - \mathbf{p}_i) \quad (17)$$

The angular velocity of the end-effector can be written as

$$\boldsymbol{\omega}_e = \boldsymbol{\omega}_b + \sum_{i=1}^n \dot{\mathbf{q}}_i \hat{\mathbf{k}}_i \quad (18)$$

The end-effector velocities can be written as

$$\begin{bmatrix} \dot{\mathbf{r}}_e \\ \boldsymbol{\omega}_e \end{bmatrix} = \begin{bmatrix} \mathbb{I}_3 & (\mathbf{r}_{eb}^\times)^T \\ O_3 & \mathbb{I}_3 \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_b \\ \boldsymbol{\omega}_b \end{bmatrix} + \begin{bmatrix} \mathbf{J}_{Te} \\ \mathbf{J}_{Re} \end{bmatrix} \dot{\mathbf{q}} \quad (19)$$

where $\mathbb{I}_3 = 3 \times 3$ identity matrix, $O_3 = 3 \times 3$ zero matrix, $\mathbf{r}_{eb} = \mathbf{r}_e - \mathbf{r}_b$, $\dot{\mathbf{q}} = [\dot{q}_1, \dot{q}_2, \dots, \dot{q}_n]^T \in \mathbb{R}^{n \times 1}$ is the joint rates vector, $\mathbf{J}_{Te} \in \mathbb{R}^{3 \times n}$ is the robot translational Jacobian matrix, $\mathbf{J}_{Re} \in \mathbb{R}^{3 \times n}$ is the robot rotational Jacobian matrix, and $\mathbf{r}^\times \in \mathbb{R}^{3 \times 3}$ is a skew-symmetric matrix. For any vector $\mathbf{r} = [r_1, r_2, r_3]^T \in \mathbb{R}^3$ the skew-symmetric matrix is defined as

$$\mathbf{r}^\times = \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix} \quad (20)$$

$$\mathbf{J}_{Te} = \begin{bmatrix} \hat{\mathbf{k}}_1 \times (\mathbf{r}_e - \mathbf{p}_1), \dots, \hat{\mathbf{k}}_n \times (\mathbf{r}_e - \mathbf{p}_n) \end{bmatrix} \quad (21)$$

$$\mathbf{J}_{Re} = \begin{bmatrix} \hat{\mathbf{k}}_1, \hat{\mathbf{k}}_2, \dots, \hat{\mathbf{k}}_n \end{bmatrix} \quad (22)$$

where $\hat{\mathbf{k}}_i \in \mathbb{R}^3$ is the unit vector of joint- i and $\mathbf{p}_i \in \mathbb{R}^3$ is the position vector of joint- i in the inertial frame. Now, Eq. (19) can be rewritten as

$$\begin{bmatrix} \dot{\mathbf{r}}_e \\ \boldsymbol{\omega}_e \end{bmatrix} = \mathbf{J}_b \begin{bmatrix} \dot{\mathbf{r}}_b \\ \boldsymbol{\omega}_b \end{bmatrix} + \mathbf{J}_m \dot{\mathbf{q}} \quad (23)$$

where $\mathbf{J}_b = \begin{bmatrix} \mathbb{I}_3 & (\mathbf{r}_{cb}^\times)^T \\ O_3 & \mathbb{I}_3 \end{bmatrix} \in \mathbb{R}^{6 \times 6}$ and $\mathbf{J}_m = \begin{bmatrix} \mathbf{J}_{Te} \\ \mathbf{J}_{Re} \end{bmatrix} \in \mathbb{R}^{6 \times n}$. Also, $\mathbf{r}_{ib} = \mathbf{r}_i - \mathbf{r}_b$. The position of the whole system's center of mass is

$$\mathbf{r}_c = \frac{1}{m_c} \left(\mathbf{r}_b m_b + \sum_{i=1}^n \mathbf{r}_i m_i \right) \quad (24)$$

where $m_c = (m_b + \sum_{i=1}^n m_i)$. The position of the whole system's center of mass with respect to the base-spacecraft center-of-mass is given by

$$\mathbf{r}_{cb} = \mathbf{r}_c - \mathbf{r}_b \quad (25)$$

The linear and angular velocity of link- i is given by

$$\begin{bmatrix} \dot{\mathbf{r}}_i \\ \boldsymbol{\omega}_i \end{bmatrix} = \begin{bmatrix} \mathbb{I}_3 & (\mathbf{r}_{ib}^\times) \\ O_3 & \mathbb{I}_3 \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_b \\ \boldsymbol{\omega}_b \end{bmatrix} + \begin{bmatrix} \mathbf{J}_{Ti} \\ \mathbf{J}_{Ri} \end{bmatrix} \dot{\mathbf{q}} \quad (26)$$

The equations of motion of the spacecraft-manipulator system are derived using the Lagrangian approach. The Lagrange function is defined as $L = T - V$, where T is the kinetic energy of the system and V is the potential energy of the system.

For the free-floating SMS, when the space robot system is in a microgravity environment far from any celestial body, the potential energy caused by gravity is negligible. Also, it is assumed that the SMS is not big enough to have significant gravity gradient potential energy. Therefore, in this case, the potential energy of the system is zero, and the Lagrangian equals the kinetic energy T :

$$L = T = \frac{1}{2} \left[(m_b \mathbf{v}_b^T \mathbf{v}_b + \boldsymbol{\omega}_b^T I_b \boldsymbol{\omega}_b) + \sum_{i=1}^n (m_i \mathbf{v}_i^T \mathbf{v}_i + \boldsymbol{\omega}_i^T I_i \boldsymbol{\omega}_i) \right] \quad (27)$$

The kinetic energy can be expressed as

$$T = \frac{1}{2} \begin{bmatrix} \mathbf{v}_b \\ \boldsymbol{\omega}_b \\ \dot{\mathbf{q}} \end{bmatrix}^T \begin{bmatrix} H_v & H_{v\omega} & H_{vm} \\ H_{v\omega}^T & H_\omega & H_{\omega m} \\ H_{vm}^T & H_{\omega m}^T & H_m \end{bmatrix} \begin{bmatrix} \mathbf{v}_b \\ \boldsymbol{\omega}_b \\ \dot{\mathbf{q}} \end{bmatrix} \quad (28)$$

The matrices $H_v \in \mathbb{R}^{3 \times 3}$, $H_{v\omega} \in \mathbb{R}^{3 \times 3}$, $H_\omega \in \mathbb{R}^{3 \times 3}$, $H_{vm} \in \mathbb{R}^{3 \times n}$, $H_{\omega m} \in \mathbb{R}^{3 \times n}$, $H_m \in \mathbb{R}^{n \times n}$.

$$H_v = m_c \mathbb{I}_3 \quad (29)$$

$$H_{v\omega} = m_c (\mathbf{r}_{cb}^\times)^T \quad (30)$$

$$H_\omega = I_b + \sum_{i=1}^n (I_i + m_i (\mathbf{r}_{ib}^\times)^T (\mathbf{r}_{ib}^\times)) \quad (31)$$

$$H_{vm} = \sum_{i=1}^n m_i \mathbf{J}_{Ti} \quad (32)$$

$$H_{\omega m} = \sum_{i=1}^n (I_i \mathbf{J}_{Ri} + m_i (\mathbf{r}_{ib}^\times) \mathbf{J}_{Ti}) \quad (33)$$

$$H_m = \sum_{i=1}^n (\mathbf{J}_{Ri}^T I_i \mathbf{J}_{Ri} + m_i \mathbf{J}_{Ti}^T \mathbf{J}_{Ti}) \quad (34)$$

where $\mathbf{J}_{Ti} \in \mathbb{R}^{3 \times n}$ and $\mathbf{J}_{Ri} \in \mathbb{R}^{3 \times n}$ are defined as

$$\mathbf{J}_{Ti} = \left[\hat{\mathbf{k}}_1 \times (\mathbf{r}_i - \mathbf{p}_1), \dots, \hat{\mathbf{k}}_i \times (\mathbf{r}_i - \mathbf{p}_i), \mathbf{0}_{3,n-i} \right] \quad \forall (1 \leq i \leq n) \quad (35)$$

$$\mathbf{J}_{Ri} = \left[\hat{\mathbf{k}}_1, \hat{\mathbf{k}}_2, \dots, \hat{\mathbf{k}}_i, \mathbf{0}_{3,n-i} \right] \quad \forall (1 \leq i \leq n) \quad (36)$$

The Lagrange equation can be written as

$$\frac{d}{dt} \left(\frac{\delta L}{\delta \dot{\Phi}} \right) - \frac{\delta L}{\delta \Phi} = \mathbf{Q} \quad (37)$$

where $\Phi \in \mathbb{R}^{(6+n) \times 1}$ is the generalized coordinate of the space manipulator system and is defined as $\Phi = [\mathbf{x}_b; \mathbf{q}]^T$.

4 Dynamic Coupling in SMS

The free-floating SMS works on the principle of conservation of momentum. Considering the external forces and torques acting on the SMS are negligible and assuming that the initial values of the angular and linear momentum are zero, the conservation equations can be written as

$$\mathbf{P} = m_b \dot{\mathbf{r}}_b + \sum_{i=1}^n m_i \dot{\mathbf{r}}_i = 0 \quad (38)$$

and

$$\mathbf{L} = (I_b \boldsymbol{\omega}_b + \mathbf{r}_b \times m_b \dot{\mathbf{r}}_b) + \sum_{i=1}^n (I_i \boldsymbol{\omega}_i + \mathbf{r}_i \times m_i \dot{\mathbf{r}}_i) = 0 \quad (39)$$

Equations (38) and (39) can be written as

$$\begin{bmatrix} \mathbf{P} \\ \mathbf{L} \end{bmatrix} = \begin{bmatrix} H_v & H_{v\omega} \\ H_{v\omega}^T & H_\omega \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_b \\ \boldsymbol{\omega}_b \end{bmatrix} + \begin{bmatrix} H_{vm} \\ H_{\omega m} \end{bmatrix} \dot{\mathbf{q}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (40)$$

Further using Eq. (40), a direct relation between the base velocities and the joint rates can be found as

$$\begin{bmatrix} \dot{\mathbf{r}}_b \\ \boldsymbol{\omega}_b \end{bmatrix} = - \begin{bmatrix} H_v & H_{v\omega} \\ H_{v\omega}^T & H_\omega \end{bmatrix}^{-1} \begin{bmatrix} H_{vm} \\ H_{\omega m} \end{bmatrix} \dot{\mathbf{q}} = C_{bm} \dot{\mathbf{q}} \quad (41)$$

where $C_{bm} = - \begin{bmatrix} H_v & H_{v\omega} \\ H_{v\omega}^T & H_\omega \end{bmatrix}^{-1} \begin{bmatrix} H_{vm} \\ H_{\omega m} \end{bmatrix} \in \mathbb{R}^{6 \times n}$ is the joint-to-base dynamic coupling matrix. Inserting the relation of Eq. (41) into Eq. (23) yields

$$\begin{bmatrix} \mathbf{v}_e \\ \boldsymbol{\omega}_e \end{bmatrix} = (\mathbf{J}_b C_{bm} + \mathbf{J}_m) \dot{\mathbf{q}} \\ \dot{\mathbf{q}} = (\mathbf{J}_b C_{bm} + \mathbf{J}_m)^\dagger \begin{bmatrix} \mathbf{v}_e \\ \boldsymbol{\omega}_e \end{bmatrix} \quad (42)$$

Finally, inserting the expression of $\dot{\mathbf{q}}$ from Eq. (42) into Eq. (41) yields

$$\begin{bmatrix} \dot{\mathbf{r}}_b \\ \boldsymbol{\omega}_b \end{bmatrix} = C_{bm} (\mathbf{J}_b C_{bm} + \mathbf{J}_m)^\dagger \begin{bmatrix} \mathbf{v}_e \\ \boldsymbol{\omega}_e \end{bmatrix} = C_{be} \begin{bmatrix} \mathbf{v}_e \\ \boldsymbol{\omega}_e \end{bmatrix} \quad (43)$$

where $C_{be} = C_{bm} (\mathbf{J}_b C_{bm} + \mathbf{J}_m)^\dagger \in \mathbb{R}^{6 \times 6}$ is the end-to-base dynamic coupling matrix.

References

- [1] Hanspeter Schaub and John L Junkins. *Analytical mechanics of space systems*. AIAA, 2003.
- [2] Yaobing Wang and Peijian Ye. *Space robotics*. Springer, 2021.
- [3] Markus Wilde, Stephen Kwok Choon, Alessio Grompone, and Marcello Romano. Equations of motion of free-floating spacecraft-manipulator systems: an engineer’s tutorial. *Frontiers in Robotics and AI*, 5:41, 2018.